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- In political science, we often study random events.

Sample Space and Classical Probability of an Event

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- In the case of rolling one fair die

$$Pr(S) = Pr(1) + Pr(2) + Pr(3) + Pr(4) + Pr(5) + Pr(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$$

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- $Pr(e) = \frac{\text{Number of Outcomes in Event}}{\text{Number of Outcomes in Sample Space}}$

Example of Classical Probability of an Event

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- $Pr(A) = \frac{4}{52} = \frac{1}{13}$
- If we think state A has a probability of .45 of winning a war against state B, how can we think of this in terms of classical probability?

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- What is the probability of rolling snake eyes with two fair die?

Example of Classical Probability of an Event

	1	2	3	4	5	6
1	1-1	1-2	1-3	1-4	1-5	1-6
2	2-1	2-2	2-3	2-4	2-5	2-6
3	3-1	3-2	3-3	3-4	3-5	3-6
4	4-1	4-2	4-3	4-4	4-5	4-6
5	5-1	5-2	5-3	5-4	5-5	5-6
6	6-1	6-2	6-3	6-4	6-5	6-6

- What is the probability of rolling snake eyes with two fair die?
- What is the probability of rolling matching numbers with two fair die?

Simple and Compound Events

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- Simple Event: Tom Cotton wins Arkansas senate race
- Compound Event: Tom Cotton wins Arkansas senate race and Mark Udall wins Colorado Senate race

Independence, Mutual Exclusivity, and Collective Exhaustivity

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 - Income scale
 - Ballot
 - Chart of the two die roll outcomes

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- If the events are mutually exclusive then the the joint probability of either event occurring is just the sum of the probabilities.

Joint Probability Example

Consider the probabilities that the Republicans will win the Senate seats in Louisiana and Arkansas as calculated by 538.

Arkansas	
Louisiana	

- If we assume that the Republicans winning these two seats are independent events, what is the probability that Republicans win both seats?

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 - What could cause this probability to be less than 1?

Conditional Probability

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- In your methods course you will encounter the linear model $y = \beta x + \epsilon$. This means that the value of y depends upon the value of the variable x and the random variable ϵ
- If you have a value of x , a value of β calculated from OLS, and know the distribution of ϵ , then you can calculate conditional probabilities, such as $Pr(y > 0.5|\beta, x) = Pr((\epsilon + \beta x > 0.5)|\beta, x)$.

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 - Say that the probability that state X and state Y go to war is .1 and that the probability that X wins the war, if the states go to war, is .4. What is the the probability that the states fight a war and that X wins the war?

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 - If the probability that a Republican candidate wins the senate seat in Arkansas is .55 and the probability that the Republican candidate wins the senate seat in Louisiana is .55 what is the probability that the Republicans win the seat in Arkansas *or* Louisiana?
 - $Pr(A \cup L) = .55 + .55 - .3025 = .7975$

General Formulas for Computing Probabilities

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Combinations

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- If there are 5 people in a class, how many combinations of pairings of birthdays are there?
 - The possible combinations are $4 + 3 + 2 + 1$ or we can use the formula $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{5!}{2!(3!)} = 10$ or we can write them out: 12, 13, 14, 15, 23, 24, 25, 34, 35, 45

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- What are the chances that person 1's birthday is included in a pair of the birthdays?
 - This probability is $\frac{4\text{pairs}}{10\text{combinations}} = .4$

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- If there are 5 people in class, how many permutations of pairings of birthdays are there?
- The possible permutations are $4 + 3 + 2 + 1 + 4 + 3 + 2 + 1$ or we can use the formula $P(n, k) = \frac{n!}{(n-k)!} = \frac{5!}{3!} = 20$ or we can write them out: 12, 13, 14, 15, 21, 31, 41, 51, 23, 24, 25, 32, 42, 52, 34, 35, 43, 53, 45, 54

Bayes' Rule

Bayes rule provides a way to to update posterior beliefs, given an event and prior beliefs.

$$Pr(B|A) = \frac{Pr(A|B)Pr(B)}{Pr(A|B)Pr(B)+Pr(A|\neg B)Pr(\neg B)}$$

A newborn places equal probability on whether the sun will rise again, by placing one white marble and one black marble in a bag. The next day the sun rises and the child puts another white marble in the bag, increasing the child's belief that the sun will rise again to two-thirds. Each morning this process repeats itself until the child has a near certain belief that the sun will rise again.

Bayes' Rule Graphically

Consider a voter who is trying to determine the competency of a political official after bad economic performance.

$$Pr(I|B) = \frac{Pr(B|I)Pr(I)}{Pr(B|I)Pr(I) + Pr(B|\neg I)Pr(\neg I)}$$

I	I B	I B	B	B					
---	-----	-----	---	---	--	--	--	--	--

$$Pr(B|I):$$

B	B	
I	IB	IB

 $= \frac{2}{3}$

$$Pr(I):$$

I	IB	IB
---	----	----

I	IB	IB	B	B					
---	----	----	---	---	--	--	--	--	--

 $= \frac{3}{10}$

$$Pr(B|\neg I):$$

B	B					
B	B					

 $= \frac{2}{7}$

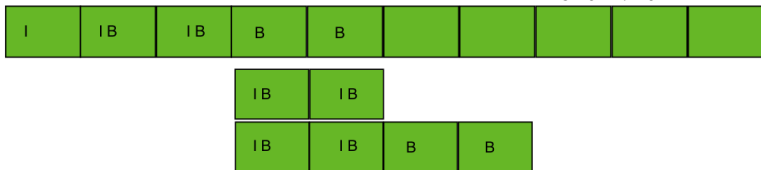
$$Pr(\neg I):$$

B	B								
I	IB	IB	B	B					

 $= \frac{7}{10}$

Bayes' Rule Graphically

$$Pr(I|B) = \frac{Pr(B|I)Pr(I)}{Pr(B|I)Pr(I) + Pr(B|\neg I)Pr(\neg I)} = \frac{\frac{2}{3} \frac{3}{10}}{\frac{2}{3} \frac{3}{10} + \frac{2}{7} \frac{7}{10}} = \frac{1}{2}$$



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	Efficiency Campaign	Security Campaign	Total
Efficiency Implementation	16	12	33
Security Implementation	0	10	10
Total	16	22	43

- What is the probability of efficiency implementation given an efficiency campaign $P(e|\epsilon)$

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- $$= \frac{\frac{16}{33} \frac{33}{43}}{\frac{16}{33} \frac{33}{43} + \frac{0}{10} \frac{10}{43}} = 1$$

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- $$Pr(s|\sigma) = \frac{Pr(\sigma|s)Pr(s)}{Pr(\sigma|s)Pr(s) + Pr(\sigma|\neg s)Pr(\neg s)}$$

- $$= \frac{\frac{10}{10} \frac{10}{43}}{\frac{10}{10} \frac{10}{43} + \frac{12}{33} \frac{33}{43}} = \frac{10}{22}$$

Bayes' Rule Example: Different Beliefs

	Efficiency Campaign	Security Campaign	Total
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- $$Pr(s|\sigma) = \frac{Pr(\sigma|s)Pr(s)}{Pr(\sigma|s)Pr(s) + Pr(\sigma|\neg s)Pr(\neg s)}$$

- $$= \frac{\frac{10}{43} \frac{10}{43}}{\frac{10}{43} \frac{10}{43} + \frac{2}{33} \frac{33}{43}} = \frac{10}{12}$$

Odds, Odds Ratio and Relative Risk

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- Odds Ratio: $\frac{\frac{Pr(x_1)}{Pr(\neg x_1)}}{\frac{Pr(x_2)}{Pr(\neg x_2)}}$

Odds, Odds Ratio and Relative Risk

- Odds of an event: $\frac{Pr(y)}{Pr(\neg y)}$
- Odds Ratio: $\frac{\frac{Pr(x_1)}{Pr(\neg x_1)}}{\frac{Pr(x_2)}{Pr(\neg x_2)}}$
- What is the interpretation, if the odds of the Broncos winning the Superbowl is 1? What is the interpretation of the odds ratio equal to 1 for the Broncos and the Seahawks winning the Superbowl?

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 - A random variable could be Y that can take on H or T from a coin toss. What would be this random variable's distribution?
 - A random variable could be the number of civilian deaths in a civil war.

Random Variables Continued

- A **realization** of a random variable is a particular value that the variable takes. In the first example, a realization of Y could be H . Often the random variable is denoted in uppercase, Y , and the realization is denoted in lower case y . Then we can use expressions like $Pr(Y > y)$.

Random Variables Continued

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- The **support** of a distribution is the set of all values that a random variable takes with a probability of greater than zero.

Sample Distributions

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- What is the relative frequency distribution?
- A **histogram** is a bar chart of the distribution of the relative frequencies in which the area under each bar is equal to the relative frequency of that value.

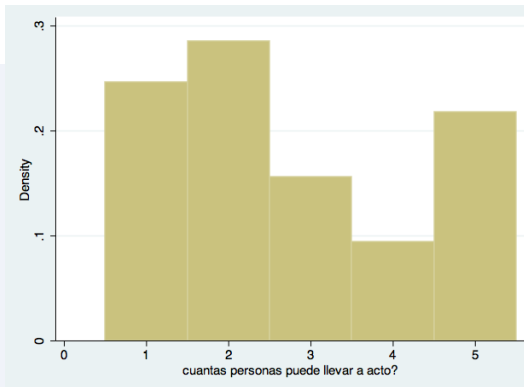
Sample Distributions

- Sample distributions are representations of the number of cases that take each of the values in a sample space for a given portion of the population.
- What is a frequency distribution?
- What is the relative frequency distribution?
- A **histogram** is a bar chart of the distribution of the relative frequencies in which the area under each bar is equal to the relative frequency of that value.
 - The formula that generates bars is $Pr(Y = y) = f_y \Delta_i$, where f_y is the height of the bar and Δ_i is the width.

Frequency Distributions: Example

tab q35

cuantas personas puede llevar a acto?	Freq.	Percent	Cum.
1	183	24.63	24.63
2	212	28.53	53.16
3	116	15.61	68.78
4	70	9.42	78.20
5	162	21.80	100.00
Total	743	100.00	



Contingency Table

	Ethnic Homogeneity	Ethnic Heterogeneity
Cooperation	Rare	Common
Conflict	Rare	Rare

```
. tab q3_alt SL
```

q3_alt	SL		Total
	0	1	
0	239	27	266
1	378	103	481
Total	617	130	747

Marginal Probabilities

The marginal probability of A is that A occurs unconditional on all other events that it may depend. Thus,

$$P(A) = \sum_{i=1}^n Pr(A|B_i)Pr(B_i)$$

Consider a voter whose likelihood of voting depends on the weather. $Pr(V|\neg R) = 0.6$, $P(V|R) = 0.5$, $P(R) = 0.3$ and $Pr(\neg R) = 0.7$. What is $P(V)$?

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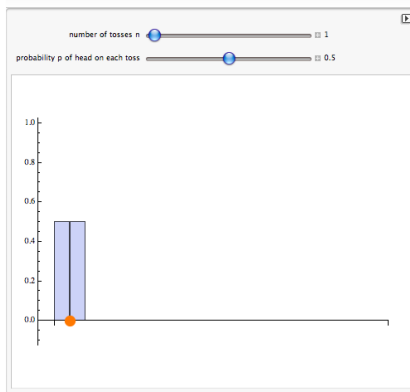
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$$\blacksquare P(V) = Pr(V|\neg R)Pr(\neg R) + P(V|R)P(R) = 0.6(0.7) + 0.4(0.3) = 0.54$$

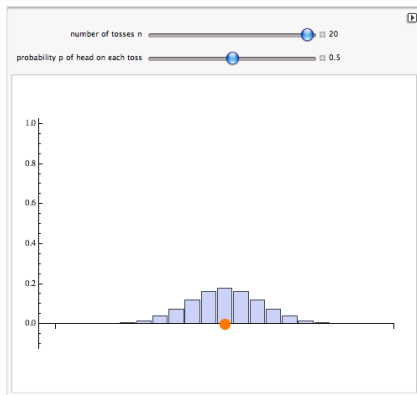
Probability Mass Function

Probability Mass Function: Let X be a discrete random variable that maps elements in a sample space to the real numbers. Then a probability mass function is a map from the set of real numbers to probabilities in $[0, 1]$: $f_X : \mathbb{R} \rightarrow [0, 1]$.



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Parameters of a PMF

A **Parameter** in PMF specifies a precise mathematical relationship among the variables in the function. Parameters can take on multiple values, and the **Parameter Space** is the set of all values that a parameter can take on. Consider any question with a binary outcome. “Is it easy for you to infer a voter’s vote choice?”

$$\begin{aligned}p(y_i = 0) &= \pi \\p(y_i = 1) &= 1 - \pi\end{aligned}$$

Here π is the parameter and $\pi \in \Theta = [0, 1]$. Θ is the parameter space.

Parameters of a PMF continued

Consider the responses to the question “Is it easy for you to infer a voter’s vote choice?”.

- We might believe that there are two groups of brokers: brokers that work 20 hours a week and brokers that work 40 hours a week. We can define the response as a binary variable that assumes a value of 0 if the broker cannot easily infer vote choice and assumes a value of 1 otherwise.

Parameters of a PMF continued

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- Then for the part time brokers it could be that $\pi = .75$ and for the full time brokers $\pi = .15$.
- So a full time broker would have a probability of .75 of easily inferring a vote choice, while a part time broker would only have a probability of .15.

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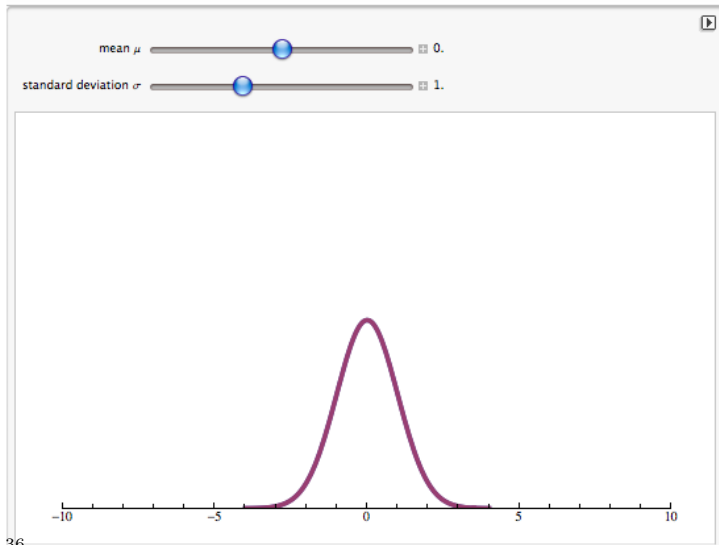
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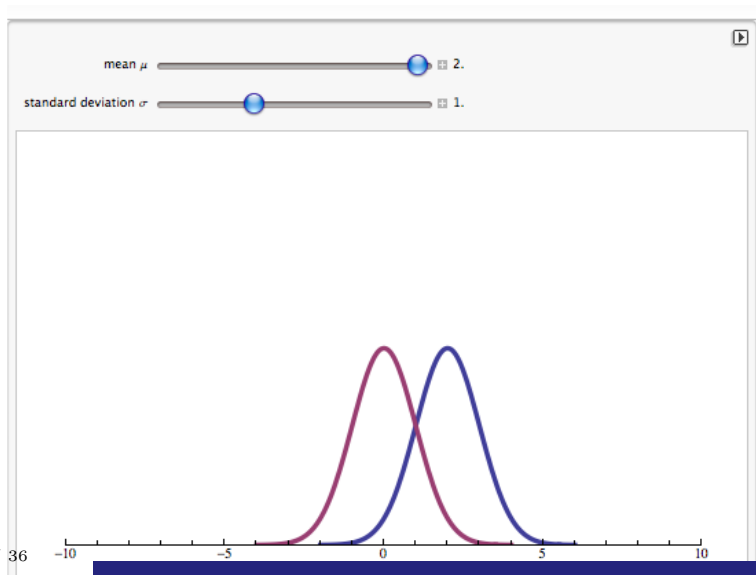
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 - In statistics this parameter corresponds to variance, often denoted σ^2

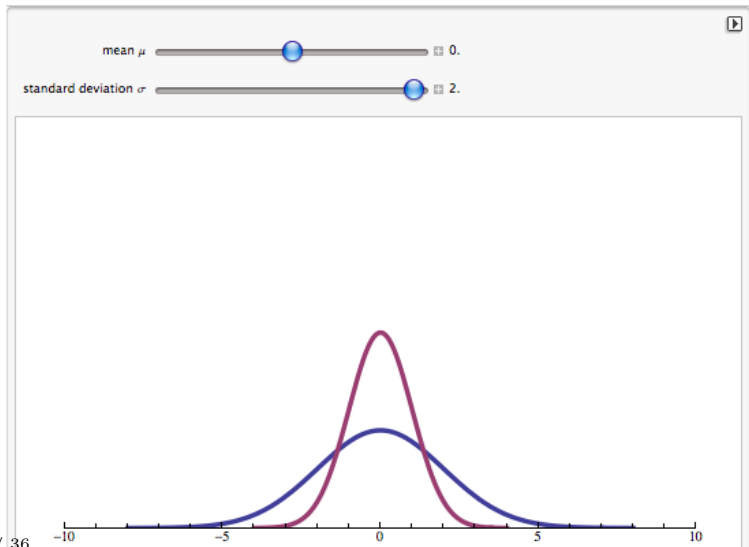
Example of the Location and Scale Parameters



Example of the Location and Scale Parameters



Example of the Location and Scale Parameters



The Cumulative Distribution Function

The **Cumulative Distribution Function** describes the function that covers a range of values below a specific value. For discrete variables the CDF can be written as:

$$Pr(Y \leq y) = \sum_{i \leq y} p(i)$$

where $0 \leq Pr(Y \leq y) \leq 1$ and $Pr(Y \leq y)$ is increasing in y .

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- If we have a PMF that describes the distribution of heights in a country, what does the CDF describe?
- If we have a PMF that describes income distribution in a country, what does the CDF describe?

CDF Graphically

